

COMS30035, Machine learning: Machine Learning Concepts

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Acknowledgement

- ▶ These slides are adapted from ones originally created by [Rui Ponte Costa](#) and later edited by Edwin Simpson.

Agenda

- ▶ The different forms of machine learning:
 - ▶ Unsupervised learning
 - ▶ Supervised learning
 - ▶ Reinforcement learning
- ▶ Other important concepts in ML:
 - ▶ Overfitting
 - ▶ Model selection
 - ▶ The curse of dimensionality
 - ▶ No free lunch theorem
 - ▶ Parametric vs non-parametric models

The different forms of machine learning



- ▶ ML attempts to learn **models** of the world
 - ▶ Usually with many simplifications
 - ▶ Models are typically a way of understanding how **input** data relates to the **outputs**
 - ▶ E.g., a function that maps weather observations to predictions

The different forms of machine learning



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 - ▶ Many different flavours of data are available!
- ▶ The type of data available, and its size, sets limits on which forms of learning we can use.

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Unsupervised learning

Only the input data is provided and models learn to extract patterns from the data.

Example tasks ¹:

- ▶ Clustering: grouping similar items together (e.g., K-means, unsupervised HMM)
- ▶ Dimensionality reduction (e.g. PCA): finding a simplified representation of input data with fewer dimensions
- ▶ Density estimation (mixture models, language models): used to estimate the probability distributions from data

¹Note that virtually all models that do not use explicit teaching signals, such as targets/labels or rewards are unsupervised.

Supervised learning

Each input is paired with an output – a “label” or “target” – and the model is trained to minimise the error (difference) between its output and the target. Sometimes we minimise the error plus a complexity penalty.

Example tasks:

- ▶ Regression: continuous outputs, typically numbers (e.g. air temperatures over time)
- ▶ Classification: discrete outputs, typically category labels (e.g. dog or bagel?)

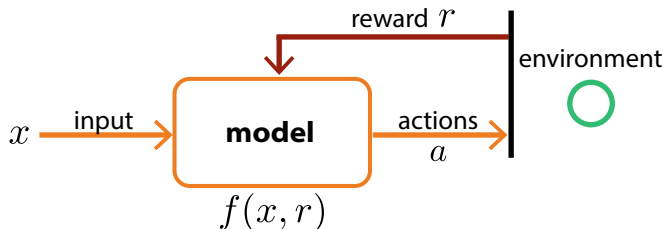
Many learning methods can be used for both regression and classification:

- ▶ Supervised neural networks
- ▶ Support vector machines
- ▶ Decision trees

Reinforcement learning

The correct outputs are not given, but the model receives a reward (or punishment) depending on its actions.

RL deals with dynamic environments where agents carry out sequences of actions. It is inspired by animal behaviour. RL is seen as a field on its own – *we do **not** teach RL on this unit.*



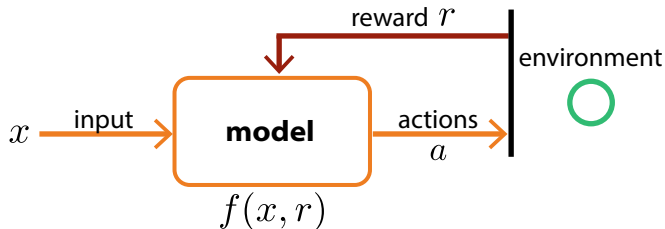
Examples:

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- ▶ Deep reinforcement learning (uses neural networks)

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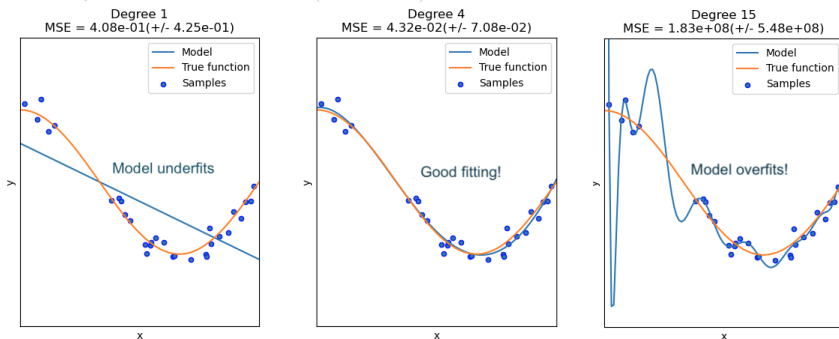
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Underfitting vs overfitting

Underfitting: A model that is too simple – it should be as “simple as possible, but no simpler.”²

Overfitting: A model that fits minor variations or noise; highly flexible models are particularly prone to overfitting.

Example from *scikit-learn* (click [here](#)):



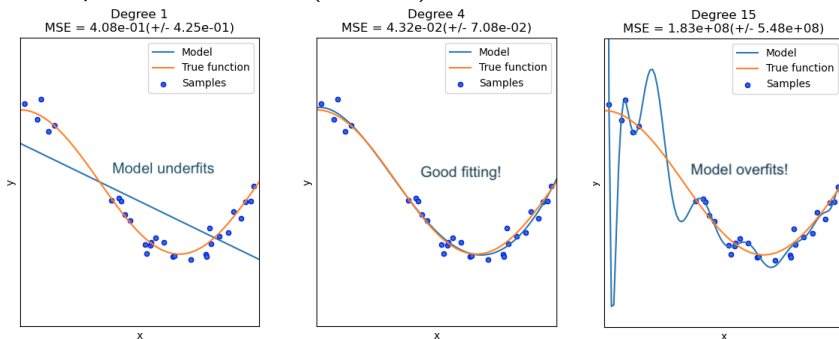
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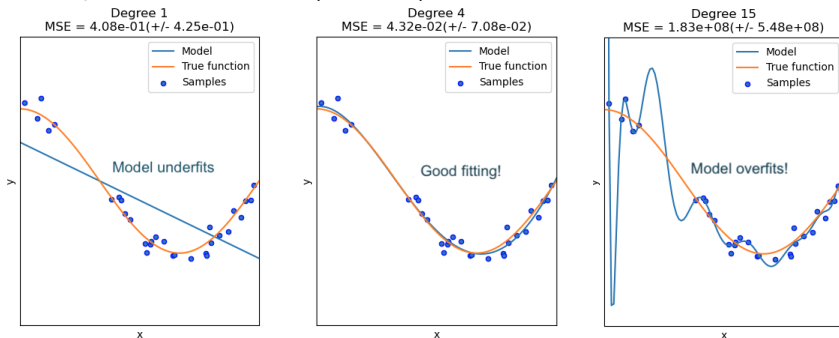
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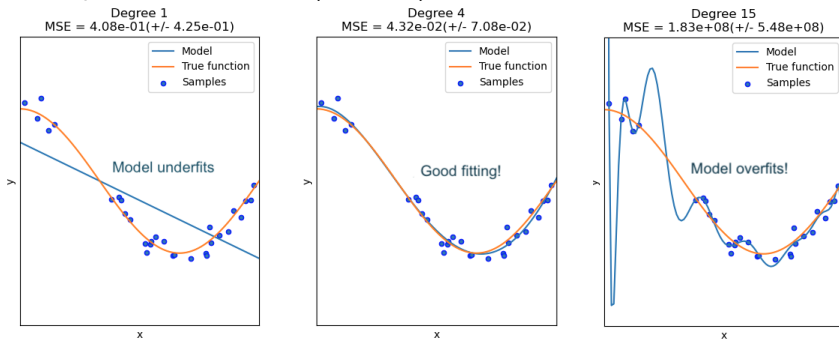
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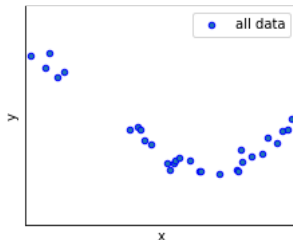
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Model selection

There are an infinite number of models, how do we choose just one?

Answer: We perform model selection to reduce under/overfitting.³

A common method is to *split the dataset*. Let's look again at the data used in the previous slide

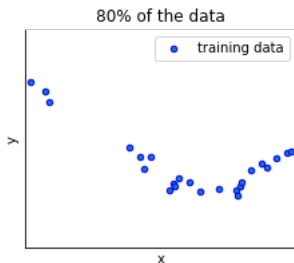


³Note that models that overfit or underfit fail to **generalise** to new data, this idea underlies model selection.

Model selection

Let's split the full dataset into:

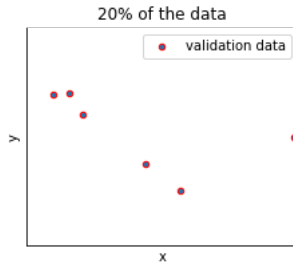
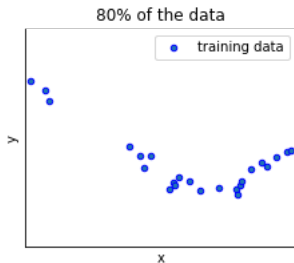
- ▶ **Training dataset:** Used for training/optimising your model (e.g. use 80% of the full dataset)
- ▶ **Validation dataset:** Used *only* for validating your model (e.g. use 20% of the full dataset)



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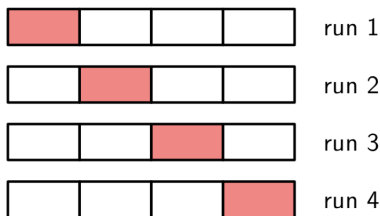
- **Testing dataset:** Used to test the model for general fitting quality after the optimisation procedure (i.e. the machine learning) has finished (e.g. use 10% of the full dataset).



Model selection

However, simply splitting the data means that we end up with less data for training the model. A solution is to cycle over multiple subsets of the data using *cross-validation*.

- **Cross-validation:** The original data is split into S groups so that $(S - 1)/S$ data is used for training. It is common to set S to a relatively low number, e.g. $S = 4$, which gives 4-fold cross validation using 3 (75% of the data) subsets for training (white blocks) and 1 for validation (red block) for each run.⁴



⁴If $S = N$ where N is the full number of data samples it gives the *leave-one-out* method.

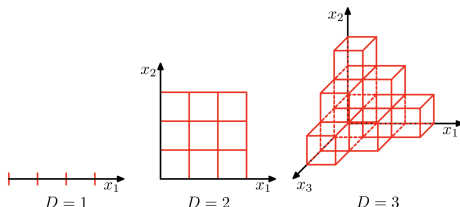
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 - ▶ **The curse of dimensionality**
 - ▶ **No free lunch theorem**
 - ▶ **Parametric vs non-parametric models**

Curse of dimensionality in ML

1D and 2D spaces can be covered by data easily, but for higher dimensions this is no longer feasible.

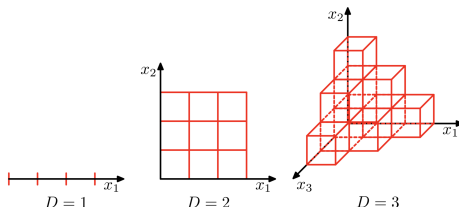
- ▶ If we were to divide the space into cells we would quickly need an exponentially large quantity of data to fill in all cells (see schematic below).
- ▶ However, its often possible to find effective algorithms for two reasons (Bishop book):
 - ▶ Data is often restricted to specific regions of the much bigger spaces – i.e. effective dimensionality is much smaller.
 - ▶ Data typically has smoothness properties – i.e. small changes in the input variables will lead to small changes in the output variables.



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No free lunch theorem

All models are wrong, but some models are useful. — George Box 1976

- ▶ Using model selection we can obtain a *good model*.
- ▶ *But* there is no universally best model selection procedure– **no free lunch theorem** (Wolpert 1996).
- ▶ Why? Models always make assumptions and these often do not generalise across domains – different domains need different models.
- ▶ No free lunch results are essentially a reinvention of the *problem of induction* studied in philosophy
- ▶ "...happenings which depend upon an infinite number of cases cannot be determined by a finite number of experiments ..."
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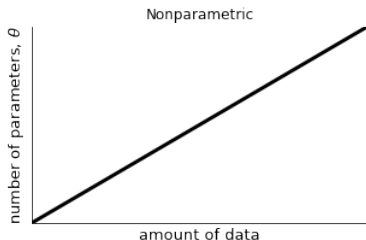
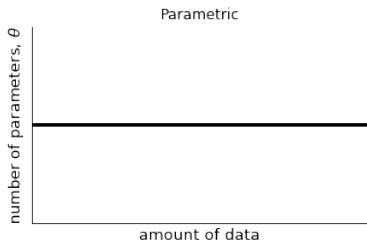
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Parametric vs non-parametric models

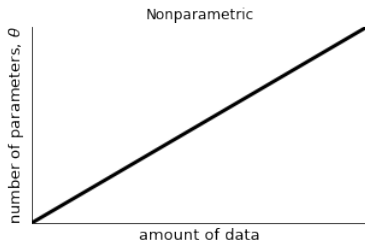
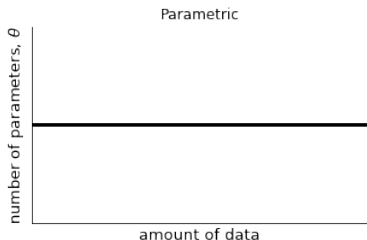
- ▶ **Parametric:** Model assumes a fixed number of parameters θ .
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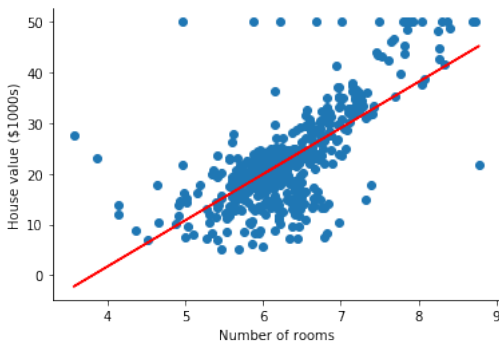


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Parametric models

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- ▶ **Cons:** Complexity is fixed. Simple models are better for simpler problems/datasets.

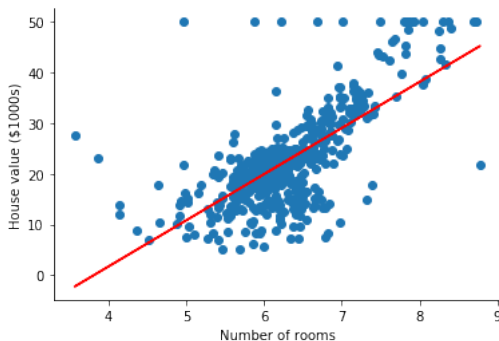
Example: Linear regression model $y = ax + b$ assumes 2 parameters $\theta = \{a, b\}$, where a is the slope and b the y-intercept.



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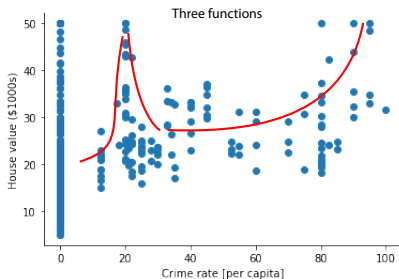
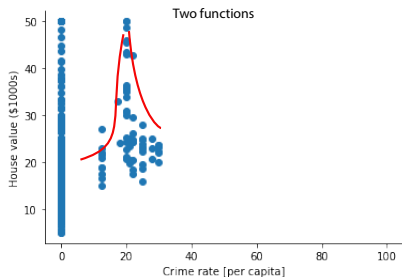
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Non-parametric models

- ▶ **Pros:** Flexible (i.e. can infer which functions to use), weak assumptions, can give better models.
- ▶ **Cons:** More expensive to train (esp. with large datasets as more parameters), risk of overfitting.

Example: Nonparametric regression with an algorithm⁶ that automatically detects which polynomial functions to use.

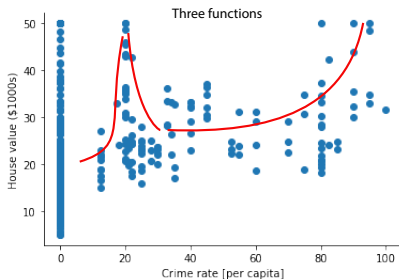
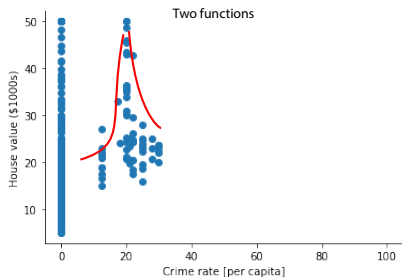


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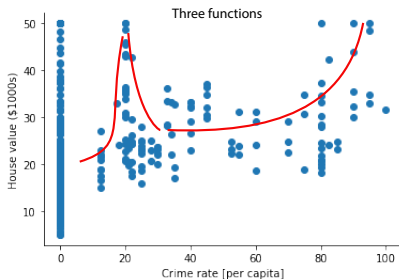
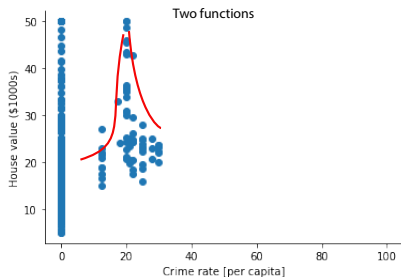


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Reading

- ▶ Bishop §1.1–§1.4. (§1.2 is about Probability Theory which we did not cover here but is useful ‘revision’.)
- ▶ The whole of Murphy Chapter 1 is a good read, but if you wish to just focus on this lecture’s material look at Murphy §1.1–§1.2.

Problems and quizzes

- ▶ No problems.
- ▶ Quizzes:
 - ▶ Week 1: Machine Learning Concepts